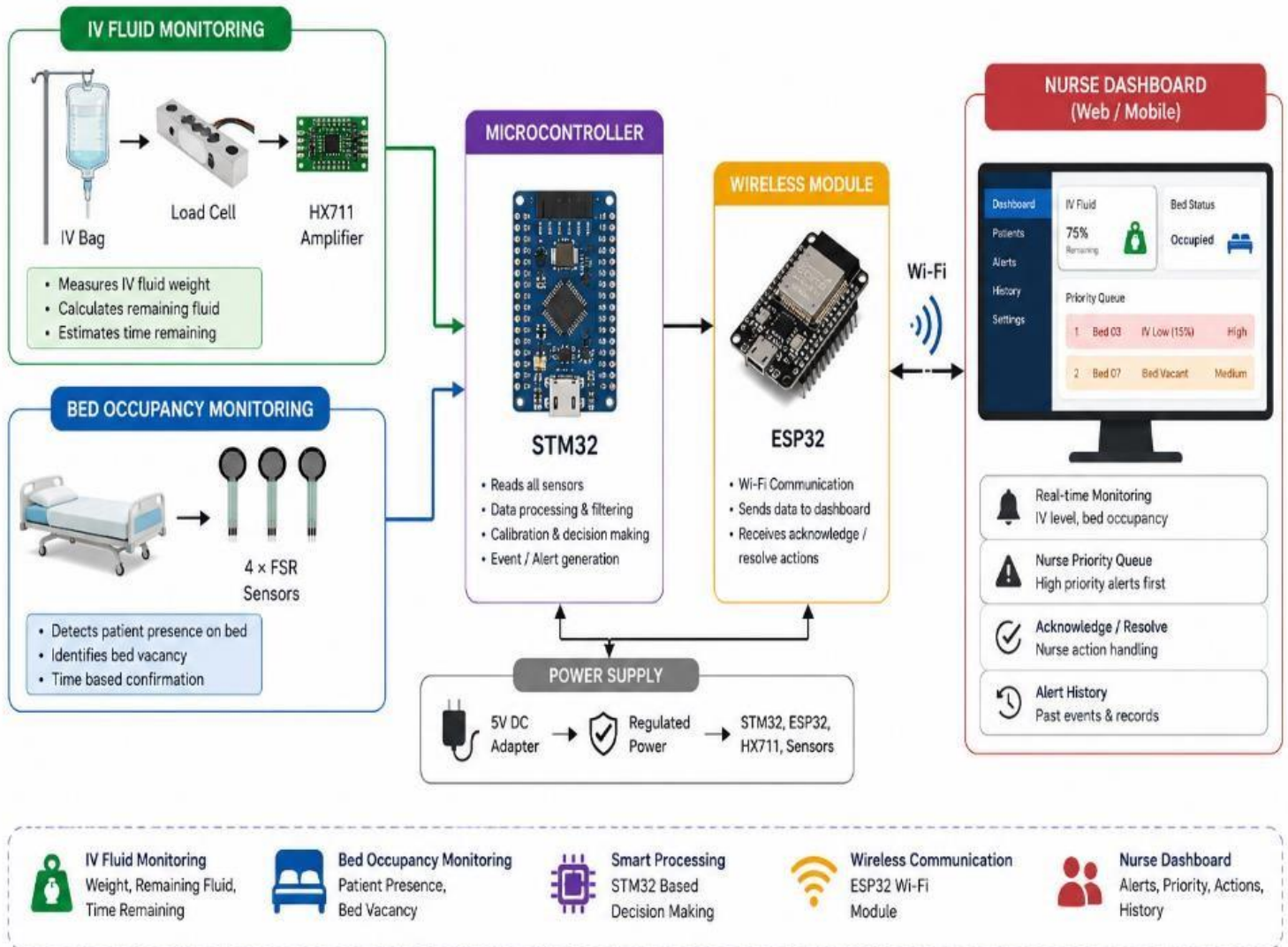


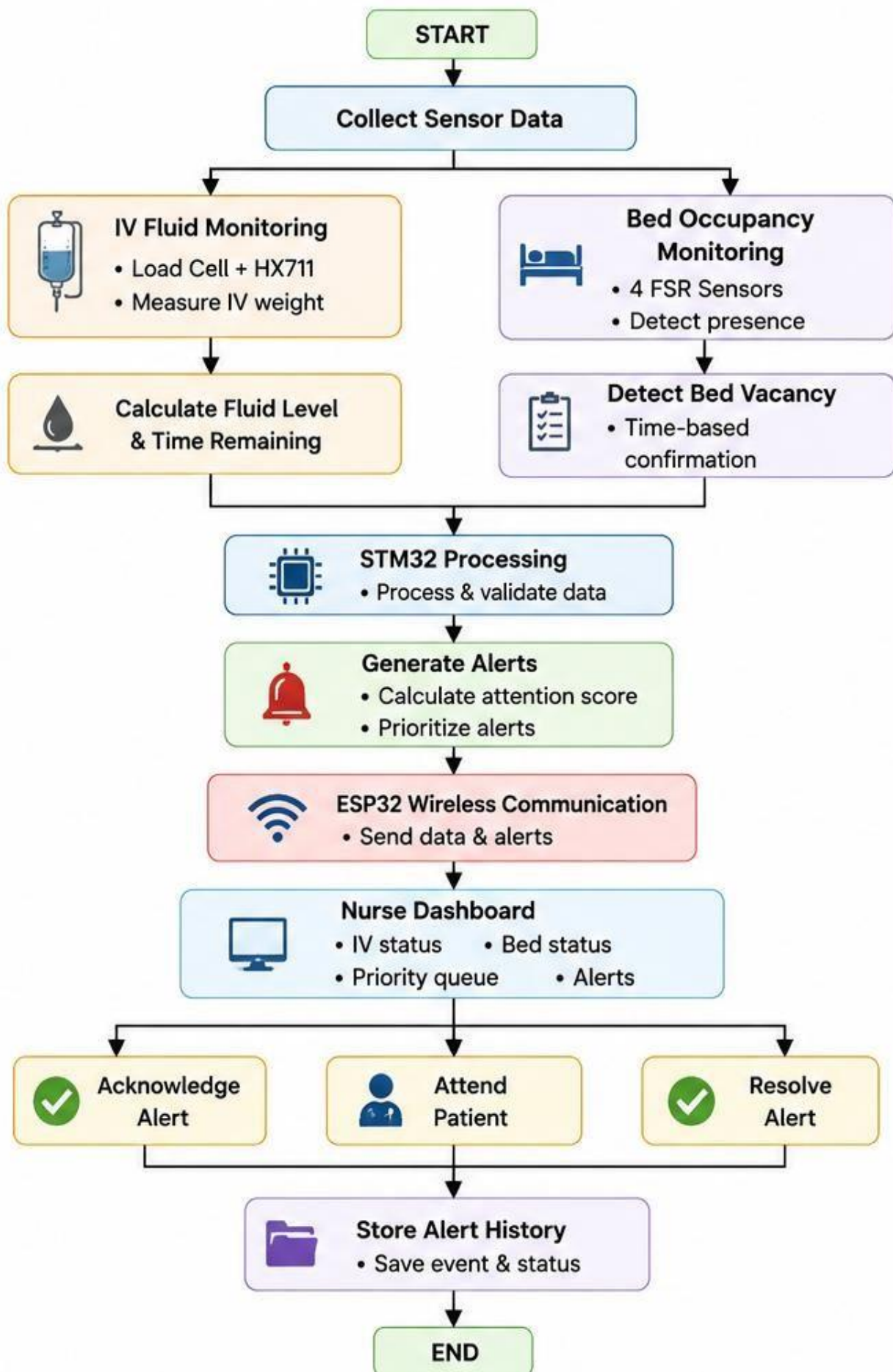
PROPOSED SOLUTION:

PROPOSED SOLUTION

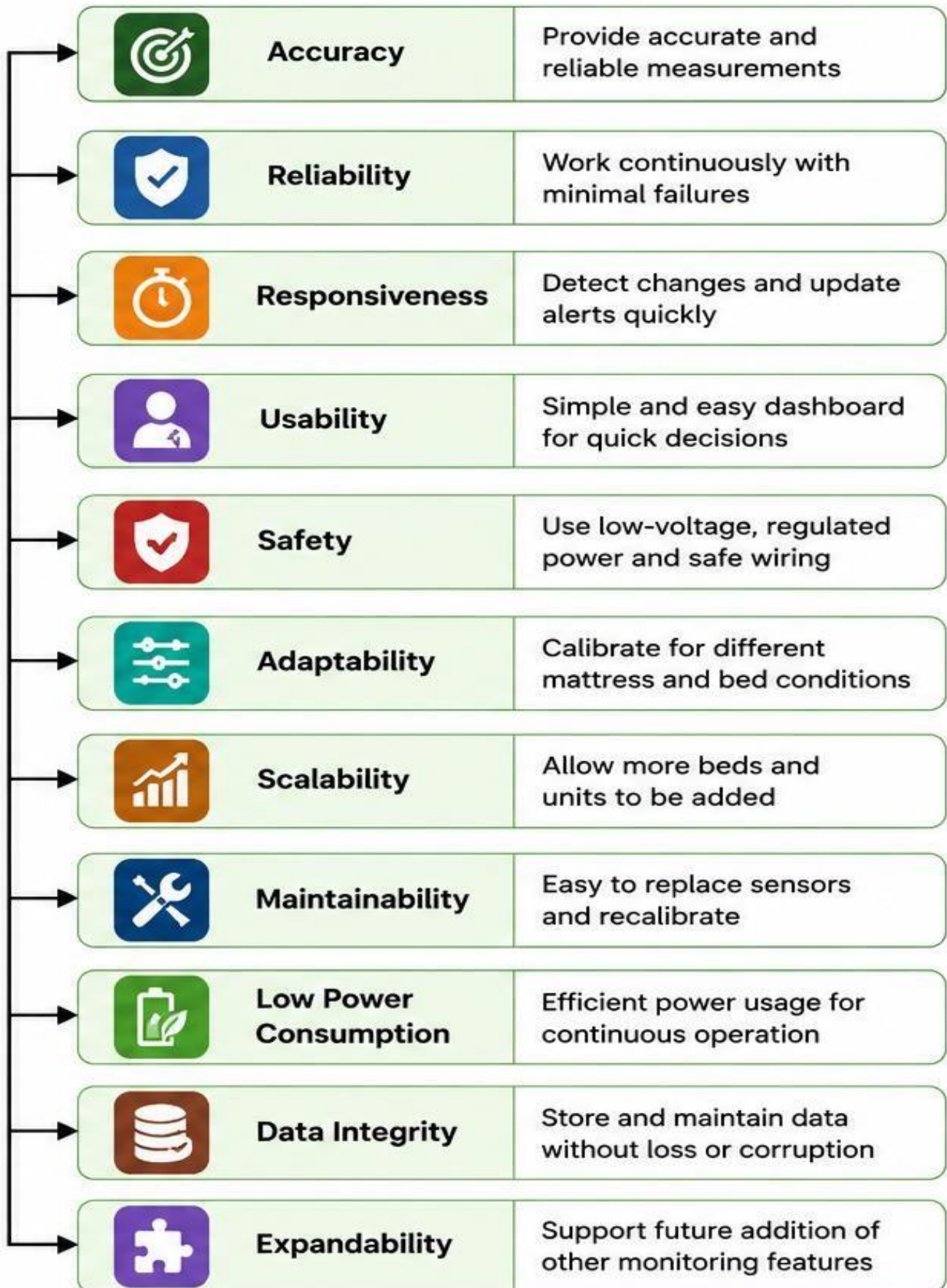
Smart Hospital Care System



Functional Requirements:



Non-Functional Requirements:



Project Planning:

Activities	Aug W1	Aug W2	Aug W3	Aug W4	Sep W1	Sep W2
Requirement Analysis						
Component Procurement						
Hardware Assembly						
Sensor Interfacing						
STM32 Programming						
ESP32 Communication						
Dashboard Development						
Predictive IV Time						
Patient Attention Score						
System Integration						
Testing & Calibration						
Documentation						

Legend:



=Planned Activity

RISK ANALYSIS

ID	RISK	POTENTIAL IMPACT	LIKELIHOOD	MITIGATION STRATEGY	CONTINGENCY PLAN
R1	 Sensor Measurement Inaccuracy May affect fluid level and alerts.	Slight error in measurement may lead to wrong estimation.	 Medium	- Proper calibration - Use quality sensors - Regular calibration check	- Recalibrate the system - Replace sensor if needed - Use safe threshold limits
R2	 Wireless Communication Issue May cause delay in alerts.	Alerts or data may be delayed temporarily.	 Medium	- Stable Wi-Fi connection - Check signal strength - Auto-reconnect in code	- Store data locally - Reconnect automatically - Alert when connection lost
R3	 System/Software Failure System may stop unexpectedly.	Monitoring may stop and some alerts may be missed.	 Low	- Watchdog timer - Stable code & testing - Regular system reset	- Auto restart using watchdog - Manual reset option - Check logs and restart
R4	 Power Failure System may turn off.	Monitoring may stop during power failure.	 Medium	- Regulated power supply - Use UPS / Power bank - Brownout protection	- Use backup power - Save data locally - Restore system safely
R5	 User Error Wrong operation or settings.	May lead to wrong actions or delayed response.	 Low	- Simple and clear UI - Proper labeling - Basic user training	- Provide confirmation prompts - Add help / guidance - Allow correction of actions

LIKELIHOOD LEVEL

 High – Likely to occur
  Medium – May occur
  Low – Unlikely to occur

Impact: Effect of risk on system performance or project.

Mitigation: Action taken to reduce or prevent the risk.

Contingency: Action taken if the risk occurs.